

MATH3171 (Mathematical Biology III)
YEAR 2022–23, PROBLEM SHEET 7 TERM

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his problem sheet is due in on Gradescope by **9am** on **Monday 21 March 2022**.

Topics covered: Chapter 5

1. **Colony Formation in an Ambush Predator-Prey System** Consider the following reaction-advection-diffusion system,

$$\begin{aligned}\frac{\partial u}{\partial t} &= u(1 - u) - uv + \nabla^2 u - \alpha \nabla \cdot (u \nabla v) \\ \frac{\partial v}{\partial t} &= -cv + uv + D \nabla^2 v,\end{aligned}\tag{1}$$

where all parameters are positive, and we have chosen a timescale such that D represents the ratio of diffusion parameters. The term involving α represents movement of the prey species u *towards* the predator, so that it is attracted to the predator. This occurs in nature for many examples of *ambush* predators, as noted in the lecture notes.

- (a) Compute the homogeneous equilibrium where both species coexist, and determine any conditions necessary for its feasibility and stability. Can this Jacobian satisfy the Turing conditions if we only had reaction and diffusion (that is, if $\alpha = 0$)?

SOLUTION: For both species to exist we need $u^* \neq 0$ and $v^* \neq 0$ so we have that an equilibrium must satisfy,

$$1 - u^* - v^* = 0, \quad -c + u^* = 0,$$

hence $u^* = c$ and $v^* = 1 - c$ is our coexistence state.

The Jacobian at this steady state is given by,

$$\mathbf{J} = \begin{pmatrix} 1 - 2u^* - v^* & -u^* \\ v^* & -c + u^* \end{pmatrix} = \begin{pmatrix} -c & -c \\ 1 - c & 0 \end{pmatrix}.$$

To have $v^* > 0$ we need $c < 1$. This is also exactly the condition we need for stability – one way to quickly see this is to note that $\text{tr}(\mathbf{J}) = -c < 0$ (which is always satisfied) and $\det(\mathbf{J}) = c(1 - c) > 0$ if $c < 1$.

As the Jacobian has a zero, it cannot satisfy the Turing conditions. Hence if there is any pattern formation in the system, we will require $\alpha \neq 0$.

- (b) In the lecture notes we derived the following two necessary conditions for diffusion-driven (Turing) instability in cross-diffusion systems:

$$d_1 G_v + d_4 F_u > d_2 G_u + d_3 F_v,$$

$$(d_1 G_v + d_4 F_u - d_2 G_u - d_3 F_v)^2 > 4(d_1 d_4 - d_2 d_3)(F_u G_v - F_v G_u).$$

Write the functions d_i for the system (1), and hence find conditions for this system to admit Turing instabilities. Simplify these conditions to find a single inequality of the form $\alpha > H(D, c)$, and briefly interpret how α and D influence the systems ability to form patterns.

SOLUTION: We have already computed the Jacobian evaluated at the steady state. If we linearize the system, using the notation from the lectures we find,

$$d_1 = 1, \quad d_2 = -\alpha u^* = -\alpha c, \quad d_3 = 0, \quad d_4 = D.$$

This leads to the following two conditions for instability:

$$-Dc > -\alpha c(1 - c), \quad (-Dc + \alpha c(1 - c))^2 > 4Dc(1 - c)$$

Simplifying these we have,

$$D < \alpha(1 - c), \quad (\alpha(1 - c) - D)^2 > \frac{4D(1 - c)}{c}$$

We can divide the first inequality by $(1 - c)$ and the second by $(1 - c)^2$ to get,

$$\frac{D}{1 - c} < \alpha, \quad \left(\alpha - \frac{D}{1 - c} \right)^2 > \frac{4D}{c(1 - c)}.$$

The first inequality can be thought of as requiring us to take the positive branch of the square root in the second. Doing this we can solve for α as,

$$\alpha > \frac{D}{1 - c} + 2\sqrt{\frac{D}{c(1 - c)}}$$

We see that, for fixed D and c , α can always be increased to lead to a spatial instability. In contrast, for fixed α and c , D cannot be increased above a threshold before the system loses the ability to pattern, though decreasing it can always allow for pattern formation if α is nonzero.

2. Cow Buckling Revisited The damped beam equation is given by,

$$\rho A \frac{\partial^2 w}{\partial t^2} + \mu \frac{\partial w}{\partial t} + \frac{\partial^4 w}{\partial x^4} + \frac{N}{EI} \frac{\partial^2 w}{\partial x^2} = 0, \quad (2)$$

where all parameters are positive and $x \in [0, L]$. This is the same equation as you saw last term, except now with a damping term μ . Instead of the clamped conditions you used last term, we will consider *free* boundary conditions of the form,

$$\frac{\partial w}{\partial x}(0, t) = 0, \quad \frac{\partial w}{\partial x}(L, t) = 0, \quad \frac{\partial^3 w}{\partial x^3}(0, t) = 0, \quad \frac{\partial^3 w}{\partial x^3}(L, t) = 0. \quad (3)$$

(a) Use the substitution $v = \partial w / \partial t$ to write eq. (2) as a first-order system in time.

SOLUTION: Making this substitution, we have the system of equations

$$\begin{aligned}\frac{\partial v}{\partial t} &= -\frac{\mu}{\rho A}v - \frac{1}{\rho A} \frac{\partial^4 w}{\partial s^4} - \frac{N}{EI\rho A} \frac{\partial^2 w}{\partial s^2}, \\ \frac{\partial w}{\partial t} &= v.\end{aligned}$$

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- (b) Using the ansatz $w = Ae^{\lambda t}f(x)$, $v = Be^{\lambda t}f(x)$, and looking for nontrivial solutions to the linear system found above, show that you get the equation,

$$(\rho A\lambda^2 + \mu\lambda)f(x) + \frac{d^4 f}{dx^4} + \frac{N}{EI} \frac{d^2 f}{dx^2} = 0, \quad (4)$$

for λ . Show that this is the same equation you would get by directly using the ansatz $w = e^{\lambda t}f(x)$ in eq. (2).

SOLUTION: Making this substitution we find the system,

$$\lambda \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} -\frac{\mu}{\rho A} & -\frac{1}{\rho A} \frac{d^4 f}{dx^4} - \frac{N}{EI\rho A} \frac{d^2 f}{dx^2} \\ 1 & 0 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix},$$

where we see that λ is an eigenvalue of the matrix on the right-hand-side. The determinant of this matrix $-\lambda I$ gives exactly eq. (4). Making the substitution directly of course gives the same answer – in a sense this change to a first order system is all to justify the use of the exponential ansatz used last term (as we know linear first-order systems can be solved by exponential functions, with some technical caveats).

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- (c) State the solutions to the eigenfunction equation,

$$\frac{d^2 g_k}{dx^2} + \rho_k g_k(x) = 0, \quad (5)$$

where the $g_k(x)$ satisfy Neumann conditions at the boundaries. What are the values of ρ_k ?

SOLUTION: We have solved this problem many times in the lectures and previous assignments, so should find,

$$g_k(x) = \cos\left(\frac{k\pi x}{L}\right), \quad \rho_k = \left(\frac{k\pi x}{L}\right)^2, \quad k = 0, 1, 2, \dots$$

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- (d) By taking $f(x) = g_k(x)$, determine an algebraic equation for the growth rates, which now depend on k so can be relabelled as λ_k . Show that these functions satisfy the boundary conditions (3).
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SOLUTION: By direct substitution of g_k into eq. (4) find,

$$(\rho A \lambda_k^2 + \mu \lambda_k) + \rho_k^2 - \frac{N}{EI} \rho_k = 0.$$

The first two boundary conditions (3) are automatically satisfied by assumption on g_k , and the latter two are a simple derivative check:

$$\frac{\partial^3 g_k}{\partial x^3}(0, t) = -\left(\frac{k\pi}{L}\right)^3 \sin(0) = 0, \quad \frac{\partial^3 w}{\partial x^3}(L, t) = -\left(\frac{k\pi}{L}\right)^3 \sin(k\pi) = 0.$$

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- (e) Determine an inequality that the model parameters must satisfy for the beam to become unstable with some spatial disturbance. Write this in terms of a critical value of the force N , such that the beam buckles for $N > N_c$.
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SOLUTION: From the quadratic equation for λ_k , we know that $\text{Re}(\lambda_k) < 0$ if all coefficients in the quadratic are positive. Hence, since $\mu > 0$ and $\rho A > 0$, for an instability we must have,

$$\rho_k^2 - \frac{N}{EI} \rho_k < 0 \implies \frac{EI k^2 \pi^2}{L^2} < N,$$

which is identical to the solution found in the notes. $k = 0$ is just a translation of the beam uniformly, so $k = 1$ is the first spatial mode. Hence the beam becomes vibrationally (spatially) unstable if the force N satisfies, $N > N_c = \frac{EI\pi^2}{L^2}$

- (f) (Optional) Compare the result you found to the critical load found for *clamped* boundary conditions in Michaelmas, given by,

$$N > N_c = \frac{EI\pi^2}{L^2}. \quad (6)$$

Hint: What does the unstable mode look like in each case?

SOLUTION: In the lecture notes, the clamped conditions led to eigenfunctions which looked like $\sin$ (under an approximation that we could neglect the derivative conditions). Hence the first buckling mode would be sinusoidal, with the beam bending outward in one direction. Our ‘free’ boundary conditions mean that the ends of the beam can move, and hence the first $\cos$ mode has the ends moving away from one another at an instability.

You can physically observe these buckling modes by holding a piece of paper tightly (clamped) and pushing inwards, which leads to a single bulge outwards. You can (sort of) see the free instability by loosely pushing it against a table, though this is harder to do without the proper setup.
